A Memory-Inspired Chatbot

Working memory, long-term memory and forgetting for dialogue

IT 207 Human Intelligence project | Team 11 | October 2026

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Problem: chatbots forget, or remember everything

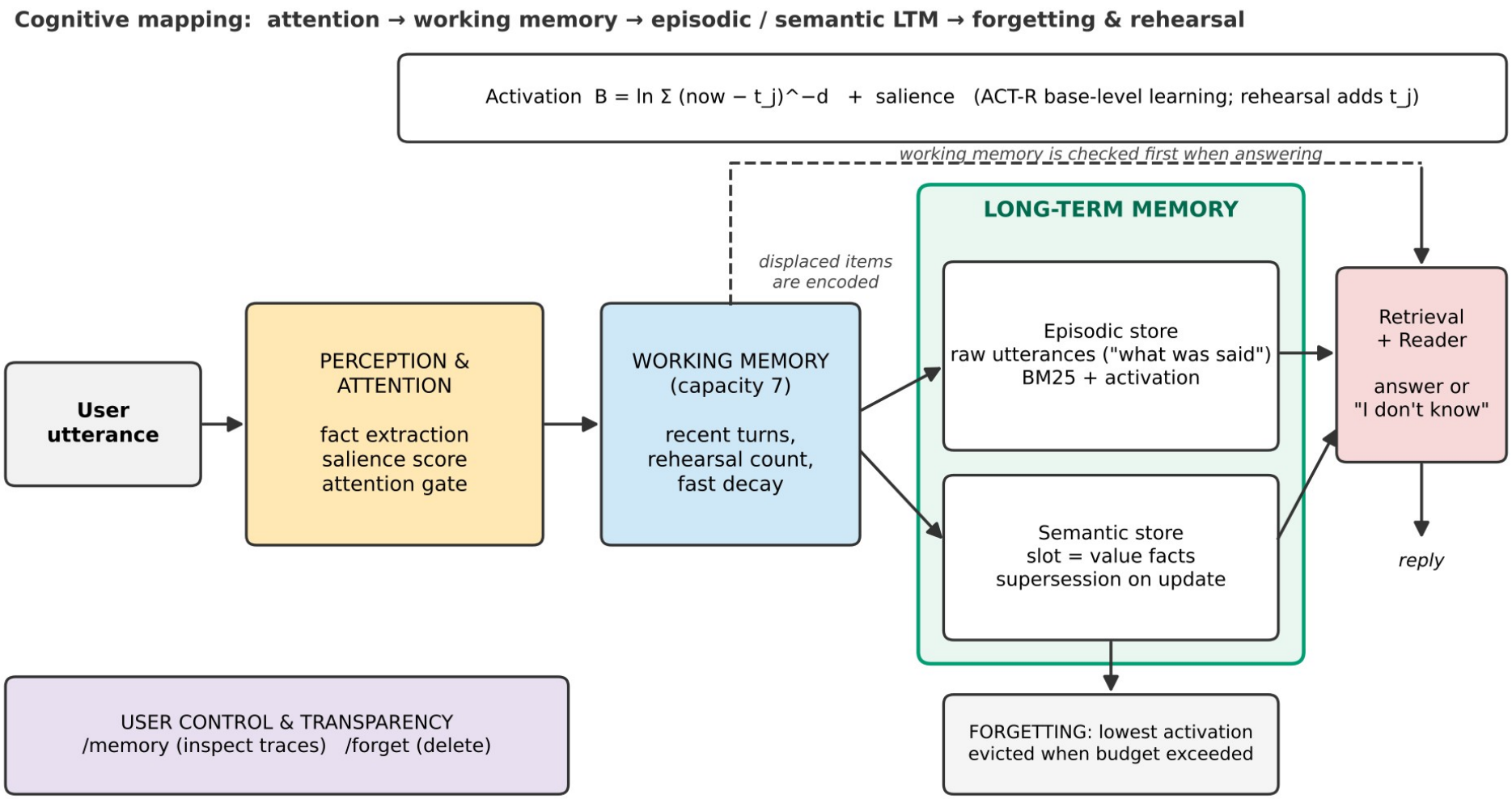
- Stateless between sessions; finite context window
- Sliding window: old facts vanish abruptly
- Store-everything RAG: unbounded growth, noisy retrieval, stale and updated facts look equally valid
- Human memory is selective, bounded and adaptive
- Question: does a memory organised like human memory recall more under the same memory budget, and which parts matter?

Computational proxy, not a measurement of anyone's intelligence; inspired by, not a model of, the brain.

Theory behind the design

- Atkinson & Shiffrin 1968: sensory, short-term, long-term stores; attention and rehearsal control transfer
- Baddeley & Hitch 1974; Miller 1956; Cowan 2001: limited-capacity working memory (~7, or ~4)
- Tulving 1972: episodic vs. semantic memory; McClelland et al. 1995: consolidation
- Ebbinghaus; Anderson & Schooler 1991; ACT-R: $B = \ln \Sigma (t - t_j)^{-d}$, decay + rehearsal

Architecture



Attention gate → working memory (7) → episodic + semantic long-term memory → activation-based forgetting under a budget.

Cognitive process → AI component

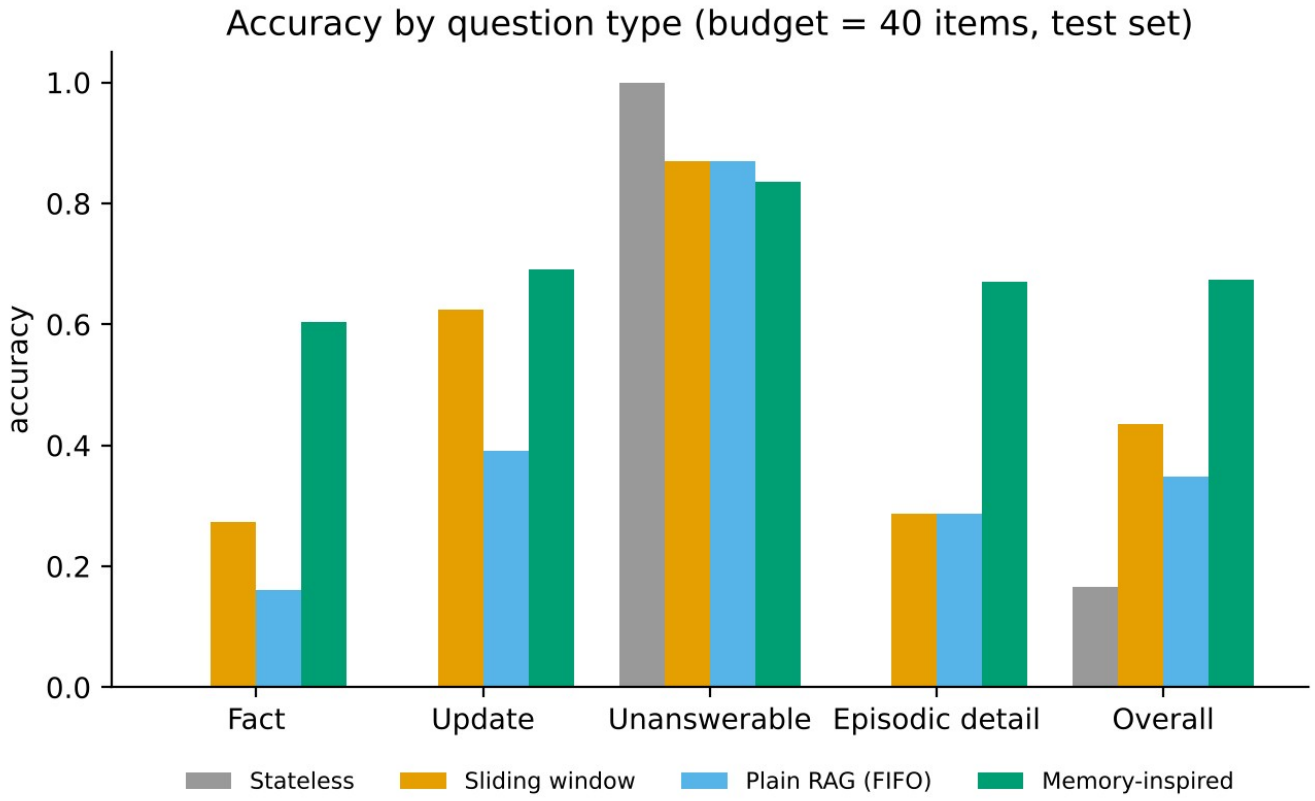
- Selective attention → salience score + attention gate
- Working memory → 7-item buffer with displacement and rehearsal counting
- Episodic memory → raw utterances + BM25 retrieval
- Semantic memory → slot = value facts; newest value supersedes the old
- Forgetting / rehearsal → ACT-R activation; lowest-activation trace evicted over budget
- User control → /memory (inspect), /forget (delete); declines to answer about third parties

Experimental design

- Synthetic multi-session personas: 5 sessions × 24 turns; facts, updates, third-party distractors, one-off episodes, chit-chat noise
- Dev templates for tuning (30 personas); held-out test templates (100 personas, 1200 questions)
- Question types: fact, update, unanswerable, episodic detail
- Baselines: stateless, sliding window, plain RAG (FIFO, k=3 and 10) at the same item budget; unbounded references
- Metrics: accuracy + 95% bootstrap CI, P/R/F1, hallucination, latency, footprint; LoCoMo evidence recall@k

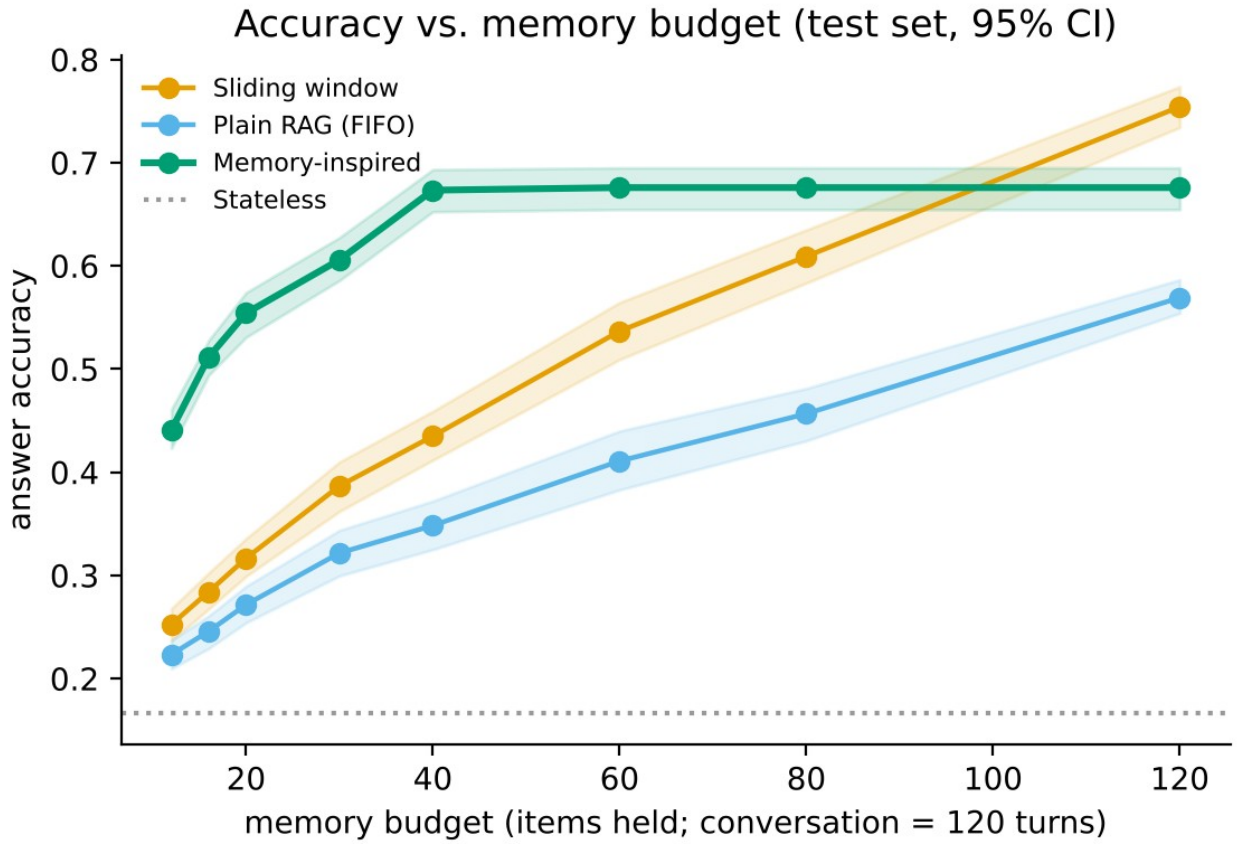
Result: 0.67 vs 0.43 vs 0.35 at budget 40

- Memory-inspired 0.673 (95% CI 0.652-0.696)
- Window 0.435; plain RAG 0.348
- Paired gain over window +0.24 (CI excludes 0)
- Full-context reference 0.754: keeping everything is better when it fits



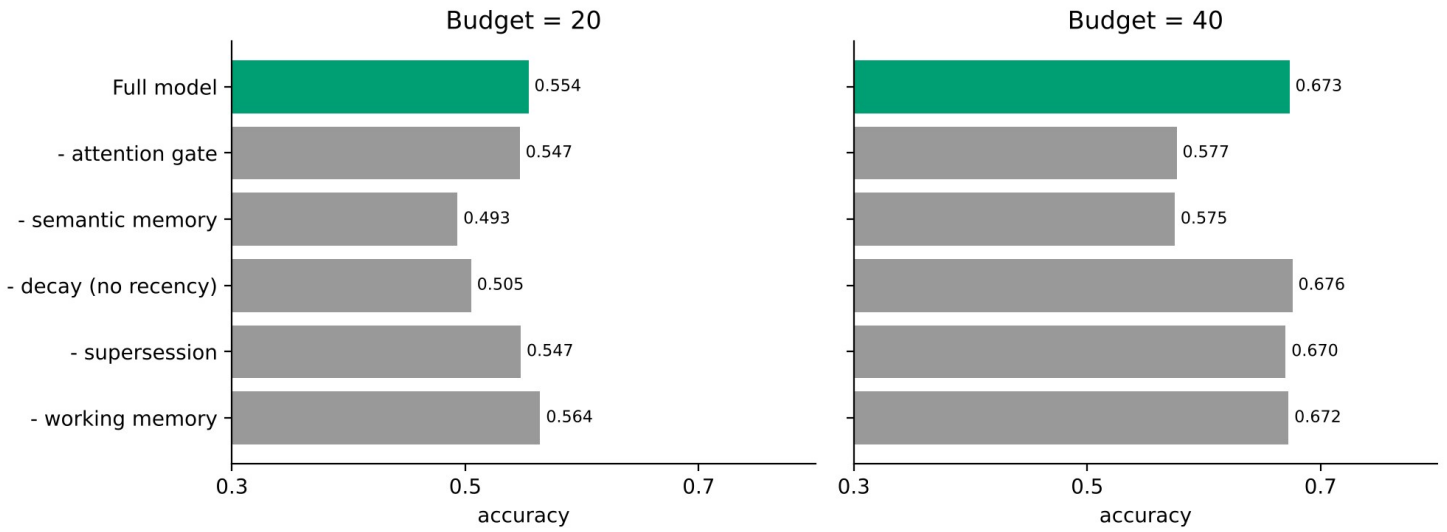
Budget sweep: help when memory is scarce

- Advantage of +0.19 to +0.24 at budgets 12-40
- Flat after 40: it has already kept what it will keep
- At budget 120 the window wins (0.754 vs 0.676)
- Claim is narrow: selective memory under scarcity



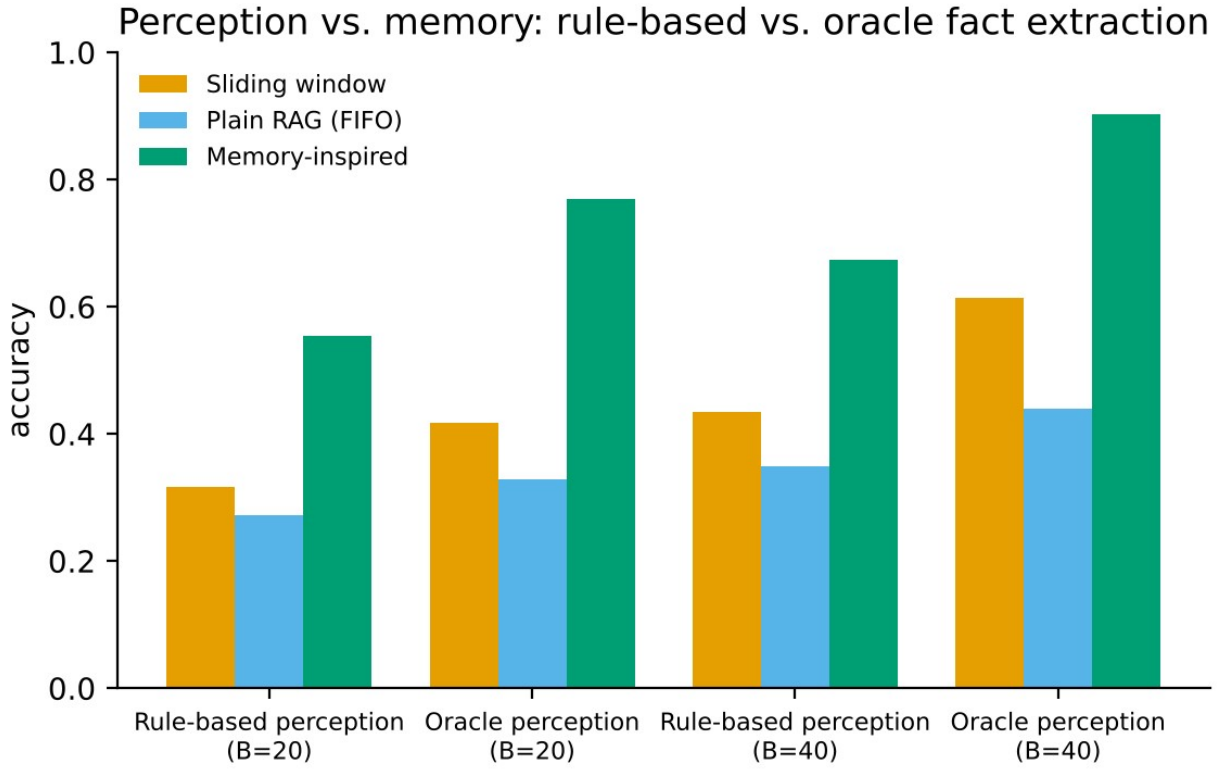
Ablations: what actually matters

- Semantic memory: biggest effect (−0.06 / −0.10)
- Attention gate: −0.10 at B=40
- Decay: −0.05 at B=20, none at B=40
- Supersession: small
- Working memory: no benefit, slightly hurts at B=20 (negative result)



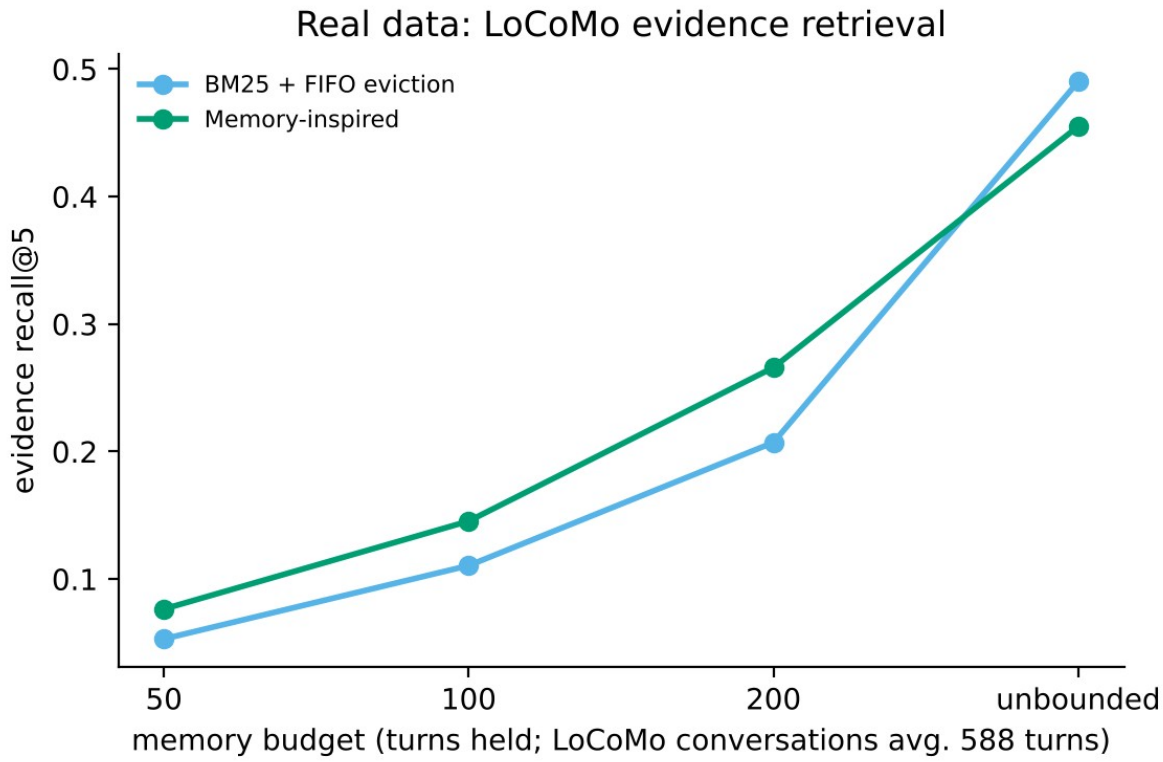
Perception vs. memory

- Extractor on held-out phrasing: P 0.96, R 0.73
- Question parser covers 79% of held-out questions
- Oracle perception: 0.90 vs 0.61 (window) at B=40
- Bottleneck = rule-based perception → use an LLM extractor



Real data: LoCoMo evidence retrieval

- 1536 questions, 10 human-reviewed conversations
- Fixed budgets: higher recall@5 than BM25+FIFO (0.145 vs 0.111 at 100 turns)
- Unbounded: BM25 is better (0.490 vs 0.455)
- Retrieval only: no reader evaluated



Live demo: persistence, update, abstention, control

- python chat.py (/memory, /forget, /skip)

you> What is my name?

bot> Your name is Priya.

you> Where do I live?

bot> You live in Toronto.

you> Where does my sister live?

bot> I only keep track of facts about you, not about other people.

you> What is my favorite color?

bot> I don't know — you haven't told me that.

you> /forget city

bot> Forgot your city (6 items removed).

Ethics and responsible use

- Privacy: bounded retention; /memory transparency; /forget deletion; no answers about third parties
- Gaps: third-party text still stored until evicted; no encryption or user separation
- Bias: English, capitalised Latin-script names; heuristic salience decides what is 'important'
- No over-claiming: not a measure of intelligence; ACT-R values borrowed, not fitted

Limitations and future work

- Synthetic templated benchmark; rule-based perception; templated replies; no human evaluation
- Working memory buffer did not help here; budget counted in items, not tokens
- Next: LLM extractor/reader; embeddings; learned salience; LongMemEval and MSC; consolidation by summarising; privacy engineering

Conclusions

- Under scarce memory, a cognitively organised memory answered 0.67 vs 0.43/0.35, and 0.90 with perfect perception
- Gain comes from compact semantic memory and selective encoding; decay matters when memory is tight
- Not better than keeping everything when everything fits
- Theory-inspired design is useful for deciding what to forget, not automatically smarter
- Code, data generator and all numbers are reproducible (see README)